The price of a post-selected signal: a sharp survival–distinguishability trade-off, saturated by broken-$\mathcal{PT}$ dynamics

Jonathan Torkelson
Embeddetech, Inc., Tulsa, Oklahoma, USA
jtorkelson@embeddetech.com

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Abstract

Local lossy (e.g. non-Hermitian) evolution on one arm of an entangled pair cannot signal; conditioning on survival famously produces an apparent signal, whose location in the post-selection is by now well understood. What has been missing is its *price*. We prove a sharp trade-off: for a qubit pair prepared in a Bell state, Alice's message encoded as a survival effect M versus its spatial flip XMX, the total-variation distance between Bob's survival-conditioned marginals obeys

$$\mathrm{TV}^2 \;+\; \Big(\frac{\sqrt{\det M}}{p}\Big)^2 \;\le\; 1,$$

where p is the survival probability. For dynamics generated by a lossy Hamiltonian, $\sqrt{\det M} = |\det C| = \exp[-\frac{1}{2}\int_0^t \mathrm{Tr}\Gamma(s)\,ds]$ is the accumulated total-loss factor—a message-independent invariant—so the conditioned signal is bounded exactly by the *anisotropy* of the loss, and anisotropy forces the total-loss factor strictly below the survival probability. The balanced gain/loss ($\mathcal{PT}$) dimer saturates the bound at every coupling and every time, in both phases: broken-$\mathcal{PT}$ dynamics traces the extremal frontier of post-selected signaling. The classical (configuration-level) signal obeys the *same* bound with a strictly smaller equality set (uncoupled loss), and the two signals close in exact quadrature, $\mathrm{TV}^2 = \mathrm{TV}_{\mathrm{cl}}^2 + \chi^2$ with χ the conditioned flip-odd coherence; a coupled broken-phase dimer therefore pays a strict classical deficit, and its classical asymptote reproduces the closed form $\tilde{\kappa}/\sin\theta$ obtained earlier as a corollary of the terminal-indivisibility theorem. We further prove a fidelity-type bound for arbitrary pure inputs and unequal survival probabilities that reduces exactly to the sharp bound at the Bell state, and a general-dimension bound via Lidskii's theorem, and we report a numerically established failure: the Bell state is *not* extremal at fixed invariants. All statements are verified numerically, including $\sim 1.8 \times 10^5$ randomized violation searches.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof, and every numerical claim is reproducible from the public scripts (see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository (see Code availability).

1 Introduction

Lee, Hsieh, Flammia, and Lee showed that local $\mathcal{PT}$-symmetric evolution on one arm of an entangled pair, treated as fundamental, permits superluminal signaling [3]. The resolution literature has located the apparent signal precisely: it was pinned experimentally to the post-selected (survival-conditioned) subensemble [4], consistency is restored in metric-corrected and embedded treatments [5, 6], and in a companion paper we reproduced the full bookkeeping in classical-probability terms and computed the signal's closed-form magnitude for the broken-$\mathcal{PT}$ dimer *at the configuration (beable) level*, $\mathrm{TV}_{\mathrm{cl}}(t \to \infty) = \tilde{\kappa}/\sin\theta$ [2]. A parallel literature quantifies what post-selection can buy in metrology: success probability trades against Fisher information [7, 8], and asymptotic post-selected hypothesis testing is governed by projective metrics [9, 10]. But the elementary single-shot question underlying the $\mathcal{PT}$ controversy appears never to have been posed as such: *how large can the conditioned signal be, given the survival probability?* (As we show in Remark 3.3, the core inequality can in fact be assembled from the Hilbert-projective-metric toolkit of Reeb, Kastoryano, and Wolf [9]—whose post-selection results are otherwise deliberately blind to the success probability—but the statement, its invariant, and its saturation structure appear nowhere.)

This paper answers it with a sharp inequality (Theorem 3.1) whose right-hand side involves one further invariant: the determinant of Alice's survival effect. Three features seem worth advertising in advance. First, the bound is *Pythagorean*: signal^2 plus (normalized total-loss factor)^2 cannot exceed one. Second, it is exactly saturated, at every instant of time, by the balanced gain/loss dimer that is the standard laboratory system of experimental $\mathcal{PT}$ optics—the dynamics at the center of the controversy is the extremal case of the phenomenon. Third, the same right-hand side bounds the *classical* (configuration-level) signal, and the two signals obey an exact Pythagorean decomposition whose third side is the conditioned coherence: a quantitative statement of how much of the post-selected anomaly is intrinsically quantum.

Everything below is stated for two-level systems first (the experimentally relevant case), then generalized. Numerical verification of every claim, including $\sim 1.8 \times 10^5$ randomized violation searches, is in the public script (see Code availability).

2 Setup

Alice and Bob share the Bell state $|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Alice applies a trace-nonincreasing operation and announces nothing; the operation's *survival effect* is M with $0 \le M \le \mathbf{1}$ (for a single-Kraus filter C, $M = C^\dagger C$; for lossy Hamiltonian evolution, $C = \mathrm{e}^{-\mathrm{i}H_{\mathrm{eff}}t}$ with $H_{\mathrm{eff}} = H - \frac{\mathrm{i}}{2}\Gamma$, $\Gamma \ge 0$). Her binary message is the *orientation* of the loss: effect $M_1 = M$ versus its spatial flip $M_2 = XMX$, $X = \sigma_x$ (the geometry of [3] and of the lossy-coupler experiments [4]). Both messages have the same survival probability

$$p = \tfrac{1}{2}\mathrm{Tr}M \qquad (\mathrm{Tr}XMX = \mathrm{Tr}M), \tag{1}$$

and the same effect determinant $\det M$. Bob's survival-conditioned marginals are $\sigma_i = M_i^T/(2p)$, and the signal is their trace distance, $\mathrm{TV} = \frac{1}{2}\left\|\sigma_1 - \sigma_2\right\|_1$—the maximal distinguishability over all of Bob's measurements. Unconditioned, Bob's marginal is exactly message-independent [2].

3 The trade-off theorem

Theorem 3.1 (survival–distinguishability trade-off)**.** *With the setup above,*

$$\mathrm{TV} \;\le\; \sqrt{1 - \Big(\frac{\sqrt{\det M}}{p}\Big)^2}, \qquad \textit{i.e.} \qquad \mathrm{TV}^2 + \Big(\frac{\sqrt{\det M}}{p}\Big)^2 \le 1, \tag{2}$$

with equality iff the eigenvectors of M are exchanged by X (equivalently: the Bloch vector of M is orthogonal to the x axis).

Proof. Let $\mu_1^2 \ge \mu_2^2$ be the eigenvalues of M, so $\mu_1^2 + \mu_2^2 = 2p$ and $\mu_1^2\mu_2^2 = \det M$. The operator $\Delta = M - XMX$ is Hermitian and traceless, so its eigenvalues are $\pm\lambda$ and $\|\Delta\|_1 = 2\lambda$. For any unit vector v,

$$v^\dagger \Delta v = v^\dagger M v - (Xv)^\dagger M (Xv) \le \mu_1^2 - \mu_2^2,$$

since $v^\dagger M v \le \mu_1^2$ and $(Xv)^\dagger M(Xv) \ge \mu_2^2$; hence $\lambda \le \mu_1^2 - \mu_2^2$. Therefore

$$\mathrm{TV} = \frac{\|M_1 - M_2\|_1}{4p} \le \frac{\mu_1^2 - \mu_2^2}{2p} = \frac{\sqrt{(\mu_1^2+\mu_2^2)^2 - 4\mu_1^2\mu_2^2}}{2p} = \sqrt{1 - \frac{\det M}{p^2}}.$$

Equality in the variational step requires $Mv = \mu_1^2 v$ and $M(Xv) = \mu_2^2(Xv)$: X maps the top eigenvector to the bottom one. In Bloch form $M = m_0\mathbf{1} + \vec{n}\cdot\vec{\sigma}$, the condition is $n_x = 0$ (X-conjugation then reverses $\vec{n}$ exactly). □

Remark 3.2 (the invariant). For Hamiltonian-generated dynamics,

$$\sqrt{\det M} = |\det C| = \exp\big[-\tfrac{1}{2}\textstyle\int_0^t \mathrm{Tr}\Gamma(s)\,ds\big]$$

is the accumulated total-loss factor (the determinant-level survival): it does not depend on the message (flipping the loss orientation preserves the trace of Γ), it is monotone in time, and it satisfies $|\det C| \le p$ always (AM–GM), with equality iff the loss is *isotropic* (C proportional to a unitary), in which case the bound forces $\mathrm{TV} = 0$. The theorem therefore says, operationally: *the conditioned signal is bounded by the anisotropy of the loss*—the normalized spread of the survival eigenvalues—*and at a fixed survival probability, anisotropy is paid for in accumulated*

loss: $|\det C|/p$ is exactly $\sqrt{1-(\text{bound})^2}$, so the total-loss factor sits strictly below p unless the loss is isotropic. For non-invertible filters (rank-deficient effects, e.g. projectors), $\det M = 0$ and the bound is vacuous: this correctly reproduces the elementary fact that static orthogonal filters achieve TV = 1 at any $p \le \frac{1}{2}$. The inequality has content precisely for dynamics.

Remark 3.3 (Hilbert-metric reading, and what is due to Reeb–Kastoryano–Wolf). The right-hand side of (2) is a projective-metric quantity in disguise: with d_H Hilbert's projective metric on the positive cone [9], $\sqrt{1-\det M/p^2} = (\mu_1^2-\mu_2^2)/(\mu_1^2+\mu_2^2) = \tanh\big[d_H(M,\mathbf{1})/2\big]$—the Hilbert distance of the survival effect from the identity. Indeed, the inequality itself can be assembled from published results of [9]: their Prop. 7 (in the arXiv version) gives $\text{TV} \le \tanh[d_H(\sigma_1,\sigma_2)/4]$, d_H is projective and transpose-invariant (so the steering map costs nothing) and X-conjugation is a d_H-isometry fixing $\mathbf{1}$, whence $d_H(\sigma_1,\sigma_2) \le 2d_H(M,\mathbf{1})$ by the triangle inequality. Their only post-selection statements, however, are probability-blind by construction (d_H mods out normalization; their Thm. 21 quantifies the success probabilities away), the steering scenario never appears, and neither does the invariant $\det M/p^2$, the saturation analysis, or the decomposition of Sec. 4. We therefore claim the statement, its operational reading, and its equality structure—not the underlying contraction technology, which is theirs (and, for the X-flipped pair, the inequality also follows from an elementary Bloch-sphere argument: both conditioned states have Bloch radius $\sqrt{1-\det M/p^2}$, and their distance is maximal when the vectors are antipodal).

3.1 Broken-$\mathcal{PT}$ dynamics traces the extremal frontier

The equality set of Theorem 3.1 is the set of effects with $n_x = 0$; what distinguishes the $\mathcal{PT}$ dimer is that it sits in this set *dynamically*, along its entire trajectory. The canonical dimer of $\mathcal{PT}$ optics is $H_{\text{eff}} = \cos\theta\,\mathbf{1} + \mathrm{i}\sin\theta\,\sigma_z + s\,\sigma_x - \mathrm{i}\sin\theta\,\mathbf{1}$ (loss frame), with $a = \sin\theta$ the loss-splitting rate, s the coupling, and exceptional point at $s = a$ [1]. Writing $c = \cosh\tilde{\kappa}t$, $h = \sinh(\tilde{\kappa}t)/\tilde{\kappa}$, $\tilde{\kappa} = \sqrt{a^2-s^2}$ (broken phase; trigonometric continuation in the unbroken phase), a two-line computation gives the survival effect in Bloch form:

$$M = \mathrm{e}^{-2at}\Big[(c^2+(a^2+s^2)h^2)\,\mathbf{1} + 2ac\,h\,\sigma_z + 2as\,h^2\,\sigma_y\Big]. \tag{3}$$

The Bloch vector lies in the y–z plane, $n_x = 0$, *at every time, every coupling, and in both phases*: by Theorem 3.1, the dimer saturates the bound identically along its whole trajectory. (Numerically: $|\text{TV} - \text{bound}| \le 1.3\times10^{-15}$ across the broken phase and $\le 4\times10^{-13}$ in the unbroken phase; a search over 99,846 random lossy generators found no violation of (2), with minimum margin 3×10^{-10}—the bound is approached arbitrarily closely but never crossed.)

In the broken phase $|\det C|/p \sim \mathrm{e}^{-2\tilde{\kappa}t} \to 0$, so the bound—and with it the quantum signal—tends to 1: deep broken-$\mathcal{PT}$ dynamics extracts the maximum possible conditioned signal permitted at its survival cost, at every instant. In the unbroken phase p oscillates at the Rabi period while $|\det C|$ decays smoothly; the signal tracks the resulting oscillating bound exactly.

4 The classical (beable) signal: same bound, different extremal family

In the stochastic-process reading of the companion papers [1, 2], Bob's physical record is his configuration, so the natural signal is the total-variation distance between the conditioned *configuration distributions*—the diagonals of $\sigma_{1,2}$: $\text{TV}_{\text{cl}} = |M_{11} - M_{22}|/(2p)$ for the X-flipped pair.

Proposition 4.1. $\mathrm{TV}_{\mathrm{cl}}$ *obeys the same bound* (2)*, with equality iff* M *is diagonal* ($n_x = n_y = 0$)*: the classical extremal family is* uncoupled anisotropic loss *(*$s = 0$*), for which* $\mathrm{TV}_{\mathrm{cl}} = \tanh(2at)$ *and* $\sqrt{\det M}/p = \mathrm{sech}(2at)$ *saturate the Pythagorean identity exactly.*

Proof. $M_{11}+M_{22} = 2p$ and $M_{11}M_{22} \geq \det M$ (the off-diagonal term only lowers the determinant), so $(M_{11} - M_{22})^2 \leq 4p^2 - 4\det M$; equality iff $M_{12} = 0$. □

The two signals are related exactly. For *every* effect and flip pair, $M - XMX = 2(n_y\sigma_y + n_z\sigma_z)$, so $\mathrm{TV} = \sqrt{n_y^2 + n_z^2}/p$ and $\mathrm{TV}_{\mathrm{cl}} = |n_z|/p$:

Corollary 4.2 (Pythagorean decomposition of the signal)**.** $\mathrm{TV}^2 = \mathrm{TV}_{\mathrm{cl}}^2 + \chi^2$*, where* $\chi = |n_y|/p = 2\,|\mathrm{Im}\langle 0|\sigma_1|1\rangle|$ *is the conditioned flip-odd coherence. (Verified to* 4.4×10^{-16} *over* 2000 *random effects.)*

Consequences. (i) $\mathrm{TV}_{\mathrm{cl}} \leq \mathrm{TV}$ always; a *coupled* dimer, whose Bloch vector has $n_y \propto ash^2 \neq 0$ for $t > 0$, saturates the quantum bound but sits strictly inside the classical one throughout the broken phase (in the unbroken phase the deficit closes at the isolated Rabi nodes $h = 0$, where signal and bound vanish together)—and by Corollary 4.2 the deficit is carried, in quadrature, exactly by the conditioned coherence χ. (ii) Along the broken dimer, $\mathrm{TV}_{\mathrm{cl}}(t \to \infty) = \tilde{\kappa}/\sin\theta = \sqrt{1 - (s/\sin\theta)^2}$, the closed form of [2], now situated: it is the classical shadow of a quantum signal that saturates its bound at 1. In the beable currency the post-selected anomaly is strictly smaller than in the quantum currency, and the gap is a measurable function of the coupling.

5 Generalizations

Proposition 5.1 (arbitrary pure inputs, unequal survival)**.** *Let the shared state be any pure two-qubit state with Bob marginal* ρ_B*, and let Alice's messages be effects* M_1, M_2 *(not necessarily flip-related) with survival probabilities* p_i*. Then*

$$\mathrm{TV} \;\leq\; \sqrt{1 - \frac{4\det\rho_B\,\sqrt{\det M_1 \det M_2}}{p_1 p_2}}. \tag{4}$$

Proof. Writing the state as ψ (a 2×2 matrix of amplitudes) with $\psi^T = \sqrt{\rho_B}\,U$ by polar decomposition, the common unitary U can be absorbed into a relabeling $\tilde{M}_i := UM_i^TU^\dagger$ of Alice's effects, which changes neither $\det M_i$ ($\det\tilde{M}_i = \det M_i$) nor the survival probabilities; without loss of generality Bob's conditioned states are $\sigma_i = \sqrt{\rho_B}\,\tilde{M}_i\sqrt{\rho_B}/p_i$ with $p_i = \mathrm{Tr}(\rho_B\tilde{M}_i)$, so $\det\sigma_i = \det\rho_B \det M_i/p_i^2$. For qubits $F(\sigma_1,\sigma_2)^2 = \mathrm{Tr}\sigma_1\sigma_2 + 2\sqrt{\det\sigma_1\det\sigma_2}$ [12], and by the rearrangement inequality $\mathrm{Tr}\sigma_1\sigma_2 \geq \lambda_1\nu_2 + \lambda_2\nu_1 \geq 2\sqrt{\det\sigma_1\det\sigma_2}$ (eigenvalues λ_i, ν_i; AM–GM), so $F^2 \geq 4\sqrt{\det\sigma_1\det\sigma_2}$; conclude with $\mathrm{TV} \leq \sqrt{1 - F^2}$ [13]. □

For the Bell state ($\det\rho_B = \frac{1}{4}$) and flip-related messages, (4) reduces *exactly* to the sharp bound (2): the general proposition interpolates to the main theorem. (Verified: 0 violations in 39,460 random draws mixing flip-related and fully independent message pairs.) We record a numerically established *negative*: the natural conjecture that the Bell state is extremal at fixed $(\sqrt{|\det C_1||\det C_2|}, \sqrt{p_1p_2})$ is **false**—randomized search finds partially entangled states exceeding the Bell envelope at those nominal invariants by as much as 0.81 (while, of course, respecting (4)); the $\det\rho_B$ dependence in (4) is real, not an artifact of the proof.

Proposition 5.2 (general dimension)**.** *Let the shared state be maximally entangled in dimension d, so that Bob's conditioned states are $\sigma_i = M_i^T/\mathrm{Tr}M$, and let the messages be $M_1 = M$ and $M_2 = GMG^\dagger$ for any unitary relabeling G. With $\mu^2_{(1)} \geq \cdots \geq \mu^2_{(d)}$ the eigenvalues of M:*

$$\mathrm{TV} \;\leq\; \frac{1}{2\mathrm{Tr}M} \sum_{i=1}^{d} \left|\mu^2_{(i)} - \mu^2_{(d+1-i)}\right|. \tag{5}$$

Proof. By Lidskii's theorem the eigenvalues of $M_1 - M_2$ are majorized by the vector $\mu^{2\downarrow} - \mu^{2\uparrow}$ of anti-sorted differences [14]; the trace norm is Schur-convex on traceless vectors, giving $\|M_1 - M_2\|_1 \leq \sum_i |\mu^2_{(i)} - \mu^2_{(d+1-i)}|$; divide by $2\mathrm{Tr}M$. □

For $d = 2$ this recovers (2) exactly. Verified: 0 violations in 2×10^4 random draws each at $d = 3, 4$, with both Haar-random and permutation relabelings (minimum margins 1.7×10^{-6}, 1.1×10^{-6}).

6 Experimental note

Both axes of the trade-off are directly measurable on existing entangled-photon platforms with tailored loss [4, 15]: p is the heralded survival rate, TV is read from Bob's conditioned tomography (or, for $\mathrm{TV}_{\mathrm{cl}}$, from his basis statistics alone), and $\sqrt{\det M}$ is fixed by the device's total loss budget. A broken-$\mathcal{PT}$ coupler swept in propagation length traces the extremal frontier $\mathrm{TV}^2 + (\sqrt{\det M}/p)^2 = 1$ point by point; any engineered deviation from the balanced dimer form (e.g. $n_x \neq 0$ loss) must fall measurably inside it. The experimental-proposal paper (`https://github.com/embeddetech/p02-measurable-ghosts`) specifies a Klauck-class device and error budget for the closely related process-level observables of [1]; the present frontier measurement can share hardware and data with it.

7 Discussion

What is and is not claimed. The bound governs survival-conditioned distinguishability; it does not challenge no-signaling, since building the conditioned ensemble requires Alice's survival record, which travels classically [4, 6, 2]. What it adds to the resolution literature is the quantitative layer: the post-selected anomaly has a hard ceiling set by loss anisotropy, the ceiling is met by the very dynamics that generated the controversy, and the classical share of the anomaly is separated from the coherent share by the same invariant. In the language of the companion program, where survival conditioning is one of exactly two entry points for apparent superluminality in a beables ontology [2], this paper prices that entry point.

Relation to prior trade-offs. Post-selected metrology bounds success-probability-weighted Fisher information [7, 8]; those are local (parameter-estimation) statements and do not imply single-shot distinguishability bounds. Asymptotic post-selected discrimination is governed by the Hilbert projective metric [9, 10], and Remark 3.3 makes the debt precise: the contraction technology is Reeb–Kastoryano–Wolf's, while the survival-probability trade-off, the invariant $\det M/p^2$, the steering formulation, and the saturation structure are new here. The optimal-filter literature (hidden nonlocality [11]) maximizes post-selected nonlocality without a survival constraint; here the survival constraint is the point.

Open questions. (i) A sharp version of (4): the correct extremal family at fixed $\det \rho_B$ is unknown (our numerics suggest skewed-marginal states with aligned filters). (ii) The multipartite and continuous-variable versions. (iii) Whether the coherence side χ of the Pythagorean decomposition (Corollary 4.2) admits its own trade-off against p. (iv) The connection, visible throughout, to the determinant bookkeeping of the divisibility analysis in [1] (there the determinant of the reduced propagator crossing zero marks the *indivisibility* onset; here $\det M$ prices the signal)—whether one statement generates both is open.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic), which carried out the technical work: the theorems and proofs, the numerical experiments, the literature analysis, and the text, including this statement. J.T. posed the research direction, provided iterative review, and takes responsibility for the decision to circulate. The research notebook in the public repository records the development, including one numerically refuted conjecture (Bell-state extremality) retained in Sec. 5.

Code availability

All results are reproducible from a single standalone NumPy/SciPy script, `scripts/t1_region_sweep.py` (∼2 minutes), in this paper's public repository

`https://github.com/embeddetech/p04-conditioned-ghosts`

which prints the saturation checks, the violation searches, the Bloch-structure verification, and all controls inline. The full running lab notebook, the companion papers, and the provenance record live in the program's hub repository, `https://github.com/embeddetech/ghosts-program`.

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